SAP-C02 Master Sheet — the 33 average questions

743 trainer questions ≈ 710 different ≈ 33 average questions. Each card: the generic question with [variable parts] → the answer pattern → clue words → traps. Reading routine: last sentence first → underline the deciding words → cross out traps.

Network

1 Connect many VPCs across many accounts

"*A company has **[number] AWS accounts** in AWS Organizations, each with **[one or more VPCs]** (CIDRs **[do not overlap]**). VPCs must **[all talk to each other / talk only within their OU or environment]** and reach **[on-premises over VPN or Direct Connect]**. New accounts are added **[every month]**. Which solution meets these requirements with the **LEAST operational overhead**?*"



Answer: One **Transit Gateway** in a network account → **shared with RAM** → attach every VPC (StackSets for new accounts) → segment with **TGW route tables**. Many Regions with one central policy → **Cloud WAN**.

Clues: hundreds of VPCs · transitive · only some may talk · new accounts

Traps: VPC peering mesh (not transitive) · VPN + third-party routing software · a TGW per account

2 Expose one service privately to other accounts

"*A company runs **[an API / a database / a SaaS service]** in a provider VPC. **[Hundreds of consumer VPCs / customers outside the organization]** must reach it **privately**; many use **[overlapping CIDRs such as 10.0.0.0/16]**. Consumers **[must not reach anything else]** and **[each connection must be approved]**. Which solution meets these requirements?*"



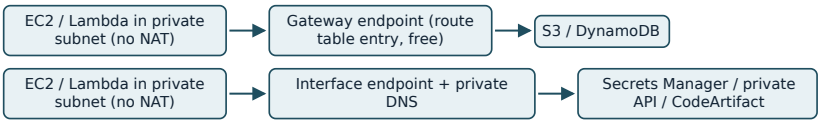
Answer: **PrivateLink**: service behind an **NLB** → **endpoint service (require acceptance, allowed principals)** → consumers create **interface endpoints**. Database with connection limits → put **RDS Proxy** behind it. Many microservices with IAM auth → **VPC Lattice**.

Clues: one service · overlapping CIDRs · approve · must not reach other resources

Traps: VPC peering or Transit Gateway (fail with overlap, expose the whole network) · NAT gateway · ALB as the endpoint service

3 Reach AWS services without the internet

"*__[EC2 instances / Lambda functions / SageMaker notebooks]__ in __private subnets with no internet access__ must use __[S3 / DynamoDB / Secrets Manager / an API Gateway API / a package repository]__. Traffic __[must not traverse the internet]__ and __[NAT gateway costs must go down]__. Which solution meets these requirements?*"



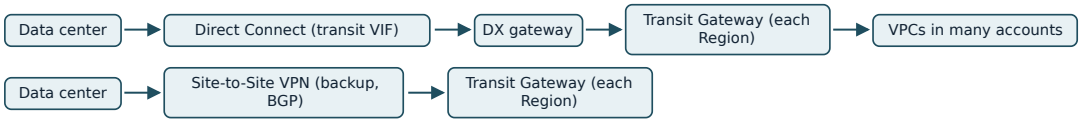
Answer: S3 / DynamoDB → **gateway endpoint** (free) · other services → **interface endpoint + private DNS** · API only from the VPC → **private API + resource policy** · packages → **CodeArtifact + endpoint**.

Clues: no internet · isolated · private subnets · NAT cost

Traps: NAT gateway (opens internet) · S3 interface endpoint when cost matters · forgetting private DNS

4 Hybrid connectivity to many VPCs

"*A company needs **[consistent, high-bandwidth, private]** connectivity from its data center to **[many VPCs in several Regions and accounts]**. If the main link fails, traffic must **[fail over automatically to an encrypted path over the internet]**. The connection **[is available in N weeks / must start now]**. Which solution meets these requirements?*"



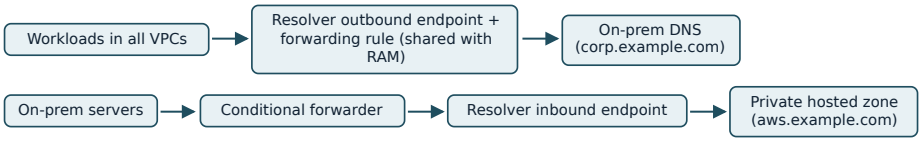
Answer: **Direct Connect → transit VIF → DX gateway → Transit Gateway**; **Site-to-Site VPN** attachment as backup (BGP prefers DX). Needs to start now → **VPN first**, DX later. Encrypt DX → **MACsec** (10/100/400G dedicated) or VPN over DX.

Clues: consistent bandwidth · many VPCs/Regions · backup path · encrypted

Traps: Private VIF per VPC · waiting weeks for new DX when it must start now · MACsec on 1 Gbps

5 Hybrid DNS

> "*Workloads in **[all VPCs]** must resolve **[corp.example.com hosted on premises]**, and on-premises servers must resolve **[Route 53 private hosted zones for aws.example.com]**. The company **[does not want to manage DNS servers]** and wants to **[minimize the number of endpoints]**. Which solution meets these requirements?*"



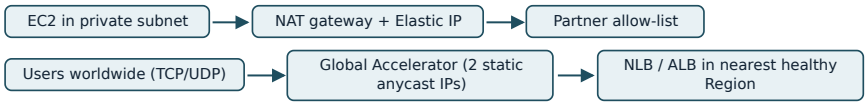
Answer: In a shared networking VPC: **outbound endpoint + forwarding rule** (shared with **RAM**, associated with every VPC) for AWS → on-prem · **inbound endpoint** + on-prem conditional forwarders for on-prem → AWS.

Clues: resolve both ways · no DNS servers · minimize endpoints

Traps: Inbound and outbound swapped · DNS servers on EC2 · endpoints in every VPC

6 Fixed IPs and global users

> "***[A partner allow-lists IP addresses / users worldwide use a UDP game / an apex domain must point to the app]**. The application runs on **[EC2 / ALB / NLB / Lambda]** in **[one or several Regions]**. Users must **[connect to the closest healthy Region / keep the same IPs]**. Which solution meets these requirements?*"



Answer: Fixed **outbound** IP → **NAT gateway + Elastic IP** · fixed **inbound** IPs → **NLB Elastic IPs** or **Global Accelerator** (also UDP, apex, fast failover) · HTTP caching → **CloudFront**.

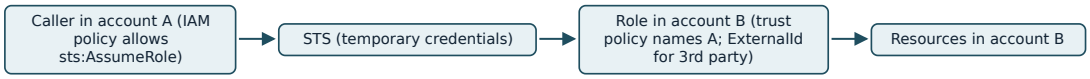
Clues: static IPs · allow-list · UDP · global · apex

Traps: ALB with Elastic IP · CloudFront for UDP or odd HTTP methods · CloudFront IPs as "static"

Identity & security

7 Cross-account access

> "***[A user / an EC2 instance / a Lambda function / an auditor / an external partner]** in **account A** must **[read / manage / stop] [resources]** in **account B**. Access must follow **least privilege** and use **[temporary credentials]**. Which combination of steps meets these requirements?*"



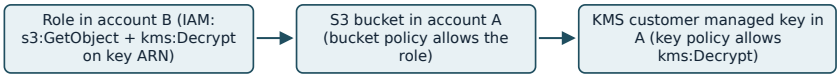
Answer: **Role in account B** with only the needed permissions; its **trust policy** names account A (or the caller role) · caller's policy allows `sts:AssumeRole` · third party → **ExternalId** · management account into members → **OrganizationAccountAccessRole**.

Clues: cross-account · least privilege · third party · temporary

Traps: SCP that "grants" · IAM users or keys per account · role created in the wrong account · trust pointing the wrong way

8 Encrypted data in another account

> "*Account B's **[Lambda role / analysts / QuickSight]** must read data in an **S3 bucket in account A** that is encrypted with **[SSE-KMS]**. Reads fail with **AccessDenied from KMS**. The solution must follow **least privilege [without copying the data]**. Which combination of steps meets these requirements?*"



Answer: Allow in **all three places**: **bucket policy** + **KMS key policy (kms:Decrypt)** + caller's **IAM policy** (S3 read + key ARN); use a **customer managed key**. Columns/rows → **Lake Formation** · huge bucket policy → **S3 Access Grants**.

Clues: SSE-KMS · AccessDenied · cross-account · columns/rows

Traps: AWS managed key (can't share) · sharing KMS keys with RAM · replicating the data

9 Workforce sign-in

> "*Employees are managed in **[Okta / Microsoft Entra ID / on-premises Active Directory]**. They must sign in to **[many AWS accounts]** with their existing credentials. Access is **by group**, users and groups must **sync automatically**, and identities must be managed **in one place**. Which solution has the **LEAST operational overhead**?*"



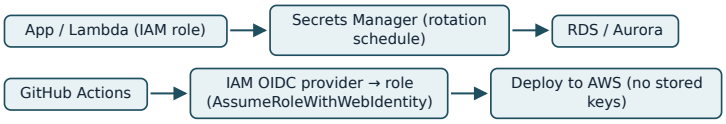
Answer: **IAM Identity Center** + external IdP over **SAML** + **SCIM** provisioning · **permission sets** assigned to groups per account · run from a **delegated admin**. On-prem AD → **AD Connector**. Other IdP for one account → **account instance**.

Clues: workforce · many accounts · sync · existing credentials

Traps: IAM users per account · Cognito (that's for customers) · Identity Center with two identity sources

10 Credentials and secrets

> "***[An application / Lambda functions / a CI/CD pipeline]*** *uses* ***[database passwords / access keys]*** *that are* ***[hard-coded / in environment variables / long-lived]***. *Credentials must* ***[rotate automatically every N days / never be stored in code]***. *Which solution has the* ***LEAST operational overhead****?*"



Answer: **Secrets Manager** with built-in **rotation** (or **IAM database authentication**) · workloads use **roles / instance profiles** · CI/CD (GitHub) → **OIDC provider + role** · SSH → **Session Manager** / **Instance Connect**.

Clues: rotate · hard-coded · long-lived · audit

Traps: Parameter Store "rotation" · KMS to store passwords · access keys in code

11 Protect the front door

> "*A public* ***[web app / API]*** *runs behind* ***[CloudFront / ALB / API Gateway]***. *It suffers* ***[SQL injection / bots / floods from changing IPs / users bypassing CloudFront]***. *Protection must* ***[not block legitimate users]*** *and* ***[log blocked requests]***. *Which solution meets these requirements?*"



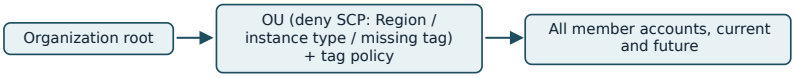
Answer: **WAF** on CloudFront/ALB/API Gateway: managed rules, **rate-based** rule, **geo** rule; start in **Count**, then Block · DDoS + response team → **Shield Advanced** · only CloudFront may reach the ALB → **CloudFront prefix list** in the SG or a **secret header** · API abuse per client → **usage plan**.

Clues: web exploits · DDoS · bypass · don't break users

Traps: Security groups to block IPs · WAF straight to Block · Shield for SQL injection

12 Organization guardrails

> "*Across* ***[all accounts / one OU]***, *nobody may* ***[use Regions other than X / launch other instance types / create resources without tags / give instances public IPs / use IAM unless admin]***. *The rule must* ***prevent*** *the action, apply to* ***future accounts***, *and* ***member admins must not be able to remove it***. *Which solution meets these requirements?*"



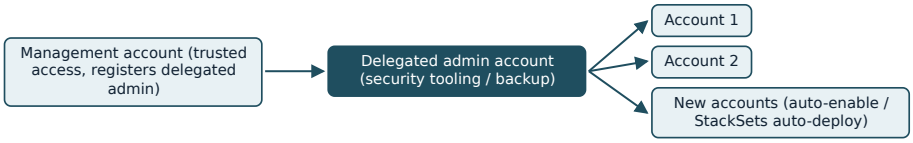
Answer: **Deny SCP with a condition** (`aws:RequestedRegion`, `ec2:InstanceType`, `aws:RequestTag`) at the root or OU · allowed tag values → **tag policy** · callers outside the org → **RCP** with `aws:PrincipalOrgID` · Control Tower → its **Region deny control**.

Clues: prevent · must not · all accounts / OU · future accounts

Traps: Config rules (detect only) · IAM policy in every account · Allow-only SCP while FullAWSAccess is attached

13 Same setup in every account

> "*Every* ***[current and future]*** *account must have* ***[an SNS topic / Config rules / billing alarms / GuardDuty and Security Hub / backup plans]***. *Member accounts* ***must not change it***, *findings must be* ***in one place***, *and the team* ***[must not use the management account]***. *Which solution has the* ***LEAST operational overhead****?*"



Answer: Resources → **StackSets** (service-managed, auto-deploy) · rules → **organization conformance pack** · new accounts → **Control Tower Account Factory** + blueprints · security services → **delegated admin + auto-enable** · backups → **backup policies**.

Clues: all accounts · future accounts · not the management account

Traps: Configure each account · custom Lambda · running daily work from the management account

Reliability

14 Pick a DR strategy

> "*A* ***[web app / database / on-premises servers]*** *runs in* ***[one Region / a data center]***. *The business needs* ***RTO [time]*** *and* ***RPO [time]*** *for a* ***[Regional outage / site loss]***. *The company wants the* ***lowest possible cost*** *and* ***[nothing running in the DR Region]***. *Which strategy meets these requirements?*"



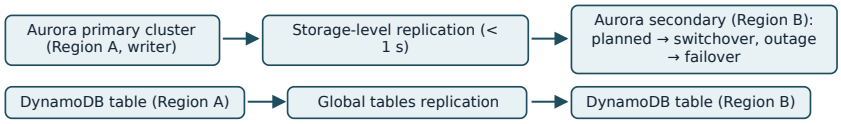
Answer: Hours/hours → **backup & restore** (AWS Backup copy + IaC) · RPO minutes, RTO hours → **pilot light** · RTO minutes → **warm standby** · ~0 → **active-active** · on-prem servers, RPO seconds → **Elastic Disaster Recovery**.

Clues: RTO/RPO numbers · lowest cost · nothing running

Traps: A more expensive pattern than the numbers need · backups too infrequent for the RPO

15 Database across Regions

> "*A **[Aurora / RDS / DynamoDB]** database must **[survive a Region loss / move to the other Region for a planned test / accept writes in several Regions]** with **[RPO of seconds / no data loss / latest value everywhere]**, **[without connection-string changes]**. Which solution meets these requirements?*"



Answer: **Aurora global database**: planned → **switchover**, outage → **failover (allow data loss)** · **DynamoDB global tables** (no loss + latest everywhere → **multi-Region strong consistency**) · cheap → **cross-Region read replica** · RDS failover <35 s with readable standbys → **Multi-AZ DB cluster**.

Clues: Region · RPO seconds · planned test · no data loss

Traps: Multi-AZ standby as DR · detach-and-promote · snapshots every 5 minutes

16 Backups that survive ransomware

> "*Backups of **[EBS / RDS / EFS]** across **[N accounts]** must be **immutable**, kept in a **separate account**, and survive **[stolen admin credentials / even the root user]**. Restores must be **[fast / into a clean account]**. Which solution meets these requirements?*"



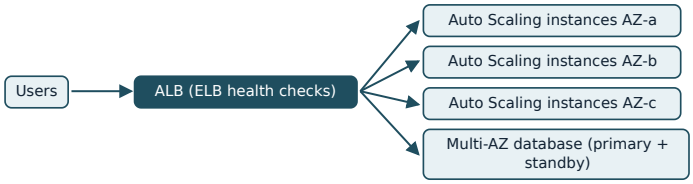
Answer: **AWS Backup** with **cross-account vault** + **Vault Lock compliance mode** + SCP protecting the vault · backup policies from a **delegated admin** · re-encrypt with a **customer managed key** · restore without copying → **logically air-gapped vault** shared via RAM.

Clues: even root · immutable · ransomware · separate account

Traps: Governance mode (removable) · SCP alone · AWS managed key · "backups in an S3 bucket"

17 Survive an AZ failure

> "*An application runs on **[a single EC2 instance / one NAT gateway / a single-AZ database]** in **[one AZ]**. It must **[stay available if an AZ fails]** and **[handle peak load without launching new instances]**. Which solution meets these requirements?*"



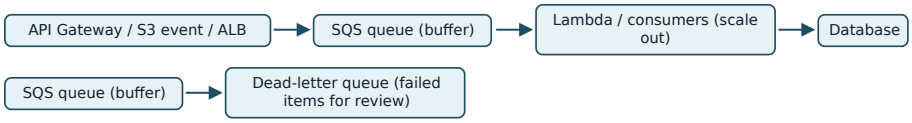
Answer: **Auto Scaling group across AZs** + ALB (ELB health checks) · **NAT gateway per AZ** (or Regional NAT gateway) · **Multi-AZ database** · static stability → enough capacity already running in the remaining AZs.

Clues: single AZ · highly available · statically stable

Traps: Bigger instance · one NAT gateway "for HA" · read replica as HA

18 Never lose data at peak

> "***[Orders / sensor data / uploads]** arrive in **[unpredictable bursts]** through **[API Gateway / S3 events / an ALB]**. Some requests **[are lost / time out]** at peak. No message may be lost, and **failed items must be kept for review**. Which solution meets these requirements?*"



Answer: **SQS between the tiers → Lambda or consumers** with a **dead-letter queue** · ordered, many readers, replay → **Kinesis Data Streams** · multi-step or human approval → **Step Functions** (task token) · fan-out → **SNS → SQS** or **EventBridge**.

Clues: spikes · never lose · retry · review failures

Traps: SNS alone (no storage) · bigger database · Lambda calling Lambda · Lambda over 15 min

19 Switch traffic away from failure

> "*Users must be sent to **[the healthy Region / the closest Region / away from an impaired AZ] [automatically / by a manual decision]**. Failover must take **[under a minute]** and **[not depend on the control plane]**. Which solution meets these requirements?*"



Answer: Route 53 **failover / latency** records with **health checks + evaluate target health** (short TTL) · fastest for web content → **CloudFront origin group** · manual, highly available → **ARC routing controls** + safety rule · bad AZ → **zonal autoshift** · orchestrated → **ARC Region switch**.

Clues: failover · closest Region · manual control · seconds

Traps: No health checks · geolocation for latency · weighted 50/50 · readiness checks (closed)

Performance & operations

20 Database overloaded

> "*At peak, **[Lambda functions / the web tier]** overload **[an RDS / Aurora database]**: **[too many connections / high read load / slow session reads]**. The fix must need **[minimal application changes]**. Which solution meets these requirements?*"



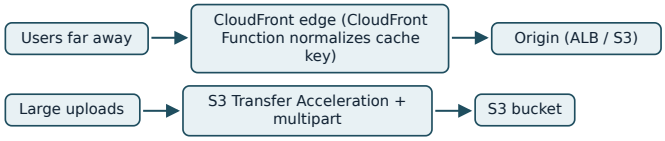
Answer: Too many connections → **RDS Proxy** (+ connect outside the Lambda handler) · read-heavy → **read replicas** / **Aurora replicas** · repeated reads or sessions → **ElastiCache** / **DAX** · write-heavy and spiky → **Aurora Serverless v2**.

Clues: too many connections · read-heavy · Lambda · minimal change

Traps: Raise max_connections · bigger instance · read replicas for writes · caching to fix writes

21 Far-away users are slow

> "*Users in **[Europe / Asia / Australia]** see slow **[page loads / downloads / uploads of large files]** from an app in **[one Region]**. The cache hit rate is **[low because of messy query strings]**. Which solution improves performance?*"



Answer: **CloudFront** in front (normalize the cache key with a **CloudFront Function** on viewer request) · large uploads → **S3 Transfer Acceleration + multipart** · APIs → **edge-optimized** endpoint or CloudFront · non-HTTP → **Global Accelerator**.

Clues: far away · slow · large files · cache misses

Traps: Lambda@Edge on origin request to raise cache hits · geolocation routing · more EC2

22 Right compute for the job

> "*A **[job / API / container / legacy app]** runs **[for N minutes or hours / at night / on demand]** and needs **[no servers to manage / OS or host control / GPUs]**. Which compute option meets these requirements with the **LEAST operational overhead**?*"

Answer: ≤15 min, event-driven → **Lambda** · longer or containerized → **ECS on Fargate** / **AWS Batch** · waits or approvals → **Step Functions** · OS/host control, custom AMI, GPUs → **EC2** / **ECS on EC2** · existing Kubernetes → **EKS**.

Clues: 15 min · hours · no servers · OS control

Traps: Lambda over 15 minutes · EC2 when "least overhead" · Fargate when host control is needed

23 Storage by protocol and access

> "***[Linux / Windows / HPC / mixed]** workloads need **[shared NFS / SMB with AD and ACLs / very high throughput / NFS + SMB + iSCSI]** storage that is **[highly available across AZs / linked to S3]**. Which storage meets these requirements?*"

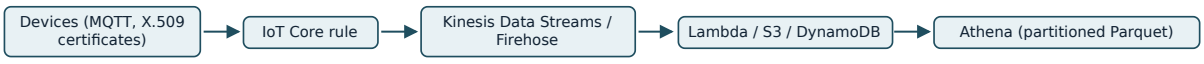
Answer: Linux shared → **EFS** · Windows SMB/AD → **FSx for Windows** · HPC → **FSx for Lustre** (+ EFA, one AZ) · multiprotocol → **FSx for NetApp ONTAP** · objects with unknown access → **S3 Intelligent-Tiering** · S3 data on EC2 → **Mountpoint for S3**.

Clues: NFS · SMB · POSIX · HPC · unknown access

Traps: EBS Multi-Attach across AZs · EFS for Windows · FSx for Lustre on Fargate

24 Streaming and IoT intake

> "***[Thousands of sensors / devices / apps]** send data **[over MQTT / in real time / in bursts]**. Data must **[not be lost]**, be **[analyzed in near real time]** and land in **[S3 / DynamoDB / a warehouse]**. Which solution meets these requirements?*"



Answer: Devices → **IoT Core** (X.509 certificates) → rule → **Kinesis Data Streams** (replay, several consumers) or **Firehose** (delivery to S3/OpenSearch/Redshift) · queries → **Athena** on partitioned Parquet.

Clues: sensors · MQTT · real time · bursts

Traps: Self-managed MQTT broker on EC2 · Firehose as a Flink source · SNS as a buffer

25 Deploy safely

> "*The team deploys **[Lambda / ECS / EC2 / Beanstalk]** changes and wants **[no downtime / 10% of traffic first / instant rollback / patched instances / data kept if a stack is deleted]**. Which approach meets these requirements?*"



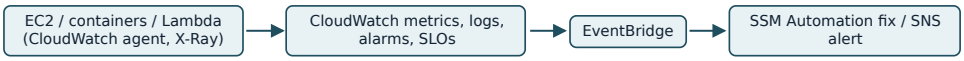
Answer: Lambda → **alias + CodeDeploy** canary/linear · ECS → built-in **blue/green, canary, linear** · Beanstalk → **swap URLs** · patched ASG → new AMI + **instance refresh** · keep data → **DeletionPolicy** Retain/Snapshot · preview → **change sets**.

Clues: canary · blue/green · rollback · patched · stack deleted

Traps: Stack policy against deletion · in-place all-at-once · Route 53 weighted for per-user stickiness

26 See and fix problems early

> "*Operations **[lose logs on scale-in / miss quota limits / can't audit SSH / find drift too late / want to prove recovery works]**. They want **[alerts and automatic fixes]** with **[no custom code]**. Which solution meets these requirements?*"



Answer: **CloudWatch agent → Logs** · **X-Ray** traces · **AWS/Usage** quota alarms · **Session Manager** · **Config rule + SSM Automation** remediation · **FIS** experiments with alarm stop · **SLOs** / **Synthetics canaries**.

Clues: logs lost · quota · SSH audit · drift · prove recovery

Traps: Custom polling Lambda · Config as prevention · CloudTrail for application logs

Cost

27 Which commitment

> "*A company spends **[amount]** on **[EC2 / Fargate / Lambda / databases]**. Usage is **[steady / steady but changing families or Regions / interruptible]** and the company will **[move to Graviton or Fargate / keep the same setup]** over **[1 / 3 years]**. Which purchase option gives the **MOST savings**?*"

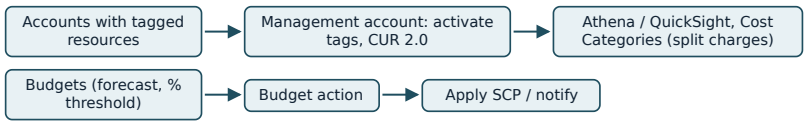
Answer: Changing compute → **Compute Savings Plan** (org level) · databases → **Database Savings Plan** · identical EC2, MOST savings → **Standard RI, 3-yr All Upfront** · interruptible → **Spot** · cover only the baseline.

Clues: steady · will change · MOST savings · interruptible

Traps: EC2 Instance SP for Fargate/Lambda · committing for peak · No Upfront for MOST savings

28 Who spends what

> "*Finance needs **[cost per team / project / business unit]** across **[N accounts]**, **[alerts at X% of a budget / automatic blocking at 100%]**, and **[past months included]**. Tags are **[inconsistent]**. Which solution meets these requirements?*"



Answer: **Activate cost allocation tags in the management account** (+ backfill) · **tag policy** + deny SCP for missing tags · **CUR 2.0 + Athena** · **Cost Categories** (split charges) · **Budgets** + budget action applying an SCP · **Cost Anomaly Detection**.

Clues: chargeback · tags · forecast · limit

Traps: Activating tags in member accounts · IAM policy as a spend limit · Trusted Advisor for cost reports

29 Stop paying for idle

> "***[Instances / clusters / tables]** are **[idle much of the day / over-provisioned / sized for peak]**. The company wants to cut cost **[without affecting production availability]**. Which solution meets these requirements?*"

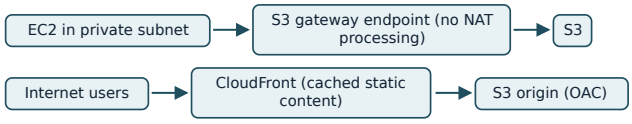
Answer: **Compute Optimizer + CloudWatch agent** (memory) → rightsize · Auto Scaling · serverless (Lambda, Fargate, **Aurora Serverless v2**, Redshift Serverless) · DynamoDB auto scaling · schedule dev instances off at night.

Clues: idle · over-provisioned · nights · peak sizing

Traps: Terminating production automatically · Spot for always-on · RIs for peak

30 Data-transfer cost

> "*The bill shows high **[NAT gateway bytes / internet data transfer out / cross-AZ traffic]** from **[EC2 reading S3 / static content / replication to on-premises]**. The fix must **[keep security and availability]**. Which solution is the **MOST cost-effective**?*"



Answer: S3/DynamoDB through NAT → **gateway endpoint** · static content → **CloudFront + S3 (OAC)** · bulk to on-prem with DX in place → **existing Direct Connect** · chatty tiers → same AZ · PrivateLink cross-AZ → AZ-local endpoint DNS.

Clues: NAT bytes · egress · cross-AZ

Traps: S3 interface endpoint · Transfer Acceleration "to save money" · Global Accelerator for cost

Migration

31 Discover and plan

> "*Before migrating **[N servers / VMware VMs / applications]**, the company needs **[an inventory / network dependencies / a TCO business case / a strategy per app / migration waves]**. **[Agents may not be installed]**. Which solution meets these requirements?*"



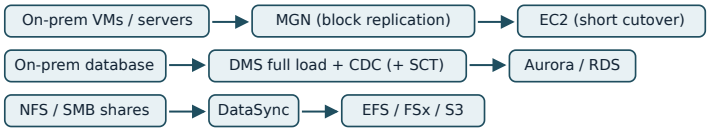
Answer: Dependencies → **AWS Transform** (exam: **Discovery Agent + Migration Hub**) · agentless utilization → collector / RVTools export · TCO → **Migration Evaluator** · per app → **retire / retain / rehost / replatform / refactor / repurchase / relocate** · pilot wave first; dependent apps move together.

Clues: dependencies · TCO · 7 Rs · waves

Traps: MGN for discovery · big-bang by a new team · moving hardware-bound apps

32 Move servers, databases or files

> "*The company must move **[VMs / an Oracle or SQL Server database / NFS or SMB file shares]** to AWS **[within N weeks]** with **[minimal downtime / no data loss]** over **[a 10 Gbps Direct Connect / the internet]**. Which approach meets these requirements?*"



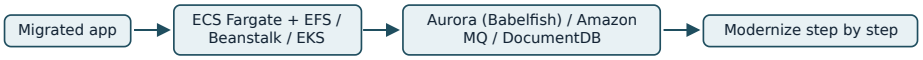
Answer: Servers → **MGN** · database → **DMS full load + CDC** (+ **SCT / Schema Conversion** for a new engine) · files → **DataSync** · on-prem keeps SMB/NFS/iSCSI → **Storage Gateway** · SFTP → **Transfer Family**.

Clues: minimal downtime · no data loss · deadline

Traps: One-time dump · Storage Gateway for a bulk copy · ordering new DX when it must start now

33 Land managed, modernize step by step

> "*After the move, the **[Java / .NET / PHP / containerized / Kubernetes]** app must be **[highly available / scalable / lower in licence cost]** with **[minimal code changes / no servers to manage]**. Which target architecture meets these requirements?*"



Answer: Same engine and protocol on managed targets: **ECS on Fargate + EFS** · **Elastic Beanstalk** · **EKS** (existing manifests) · **Aurora PostgreSQL / Babelfish** (cut licences) · **Amazon MQ** (AMQP/JMS) · **DocumentDB** (MongoDB) · modernize later in steps.

Clues: no code change · managed · licences · step by step

Traps: Full rewrite on a short timeline · Lambda for long jobs · DynamoDB for relational data